

Tom Stesco

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LINKS

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PROGRAMMING

EXPERT

Python

PROFICIENT

C++ • SQL • Bash

PRIOR EXPERIENCE

Java • Javascript

TOOLS

Airflow • Spark • PySpark
 Pytorch • PyMC3 • XGBoost
 Numpy • Pandas • SKlearn
 MySQL • BigQuery • Datastore
 Kafka • PubSub • Protobuf
 Docker • Kubernetes • Terraform
 GCP • AWS • Azure
 GNU/Linux • Git

EDUCATION

ETH ZURICH

MSC INTEGRATED BUILDING
 SYSTEMS

October 2018 | Zurich, Switzerland

UNIVERSITY OF WATERLOO

BASC HONOURS CIVIL ENGINEERING
 June 2016 | Waterloo, ON

COURSEWORK

GRADUATE

Model Predictive Control
 Mathematical Optimization
 Computational Physics
 Building Control and Automation
 Computational Fluid Dynamics
 Renewable Energy Technologies
 Technology and Innovation Management
 Innovation Leadership
 Social Networks Research

UNDERGRADUATE

Software Engineering Design Project
 Building Science
 Bioprocess Engineering
 Advanced Calculus
 Probability and Statistics
 Linear Algebra

EXPERIENCE

BCG X | SENIOR AI/ML SOFTWARE ENGINEER

May 2022 – present | Toronto, ON

- Development of **ETL and XGBoost model training** pipelines using **Spark** on **Airflow**.
- Led **ML Ops** for production ML pricing models in \$10 billion/yr ecommerce system, including: code review (team of 5), test planning and validation, CI/CD, release management, monitoring, and on-call.
- Aligned client and internal teams on design of steaming backend system using Azure Event Hubs (PaaS Kafka) and Databricks (PaaS Spark).
- Reviewed deep learning literature and evaluted published computer vision models using PyTorch.

ECOBEE | SENIOR DATA SCIENTIST

Jun 2021 – May 2022 (1 year) | Toronto, ON

- Led development of edge IoT device **convex optimization of ML models** for Model Predictive Control application in C++ using interior point methods.
- Accelerated ML modeling iteration cycle time from weeks to days by automating: preprocessing, training, validation, and deployment using Python and Docker.
- Led **external research collaborations** with 4+ academic groups and 2 industry partnership research projects on ML driven controls and simulation software.

ECOBEE | DATA SCIENTIST

Oct 2018 – Jun 2021 (2 years, 8 months) | Toronto, ON

- Scaled internal IoT **experimentation platform** from 0 to 150k+ devices running 10+ experiments simultaneously by automating build, deployment, monitoring, and metrics with C++, Python, PubSub, and Protobuf.
- Led development of **ML driven controls simulation platform**, including open sourcing **building-controls-simulator**, a Python library for HVAC control algorithm performance testing and presented at the **eSim2020** conference.

ENERGY PROFILES LIMITED | ENERGY DATA ANALYST (CO-OP)

Sep 2015 – Dec 2015 (4 months) | Toronto, ON

- Developed **SQL error detection model** for equipment issue and input data for building energy submeter data.
- Added features and bug fixes to energy bill **data ingestion pipeline** in Java.

AECOM CANADA | CIVIL ENGINEERING STUDENT (CO-OP)

Jan 2015 – Apr 2015 (4 months) | Markham, ON

- **Digitized workflow** for water flow sensor team to remove toil of: data entry from paper forms, data validation, and generating reports for clients.

DRAGADOS CANADA | CIVIL ENGINEERING STUDENT (CO-OP)

May 2014 – Aug 2014 (4 months) | Toronto, ON

- **Automated cost estimating and reporting** of road and sidewalk closures for Eglinton Crosstown LRT project bid.

UNION PEARSON EXPRESS | ENGINEERING ANALYST (CO-OP)

Jan 2013 – Apr 2013, Sept 2013 – Dec 2013 (8 months) | Toronto, ON

- Compiled data and consultant reports for **predictive modelling** of effects of pricing on ridership and revenue.

RESEARCH

AUTOMATIC CONTROL LAB | MASTER'S STUDENT

Sep 2017 – Oct 2018 | Zurich, Switzerland

- Worked with Prof. Dr. John Lygeros and Dr. Annika Eichler on Human-in-the-Loop building controls project combining Model Predictive Control (MPC) and online Bayesian learning for minimizing energy consumption and maximizing thermal comfort of occupants.
- Built Python library to model building thermodynamics and run coupled model training and simulation experiments on HPC cluster.
- Developed online Bayesian Linear Discriminant Analysis (LDA) classifier for thermal comfort using PyMC3 with an MCMC algorithm.
- Derived predictive model of occupancy probability in Python using Markov chains and change point detection algorithm.

AWARDS & PROGRAMS

2016	Excellence Scholarship & Opportunity Program	ETH Zurich
2016	Engineer of the Future Fund	University of Waterloo
2012	Merit Scholarship	University of Waterloo
2010	SHAD Valley summer program	SHAD Canada